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The influence the uncut chip thickness has on the stagnation point in orthogonal cutting

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Abstract

The effect of cutting data on the stagnation zone of a machining operation is of great interest since it governs the material flow around the cutting edge. The material flow has a significant influence on the mechanical properties of the machined surface. This paper presents a numerical model that is able to determine the effect that the uncut chip thickness has on the stagnation zone and the connection between the stagnation zone and the deformation layer in the machined subsurface.

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Keywords: Machining; Finite element method (FEM); Stainless steel; Stagnation zone; Subsurface deformation

1. Introduction

The AISI 316L alloy is used for a wide range of industrial applications where steels of type AISI 304/304L have insufficient corrosion resistance. Typical examples are: machined parts for tube fittings, valves, components for pumps and heat exchangers. It is also a commonly used alloy in production of pharmaceuticals and foods. A machining operation causes a subsurface deformation in the machined surface, insight on the surface integrity of the workpiece is essential to predict the life time of the produced part. The subsurface deformation can have a drastic effect on the performance, such as corrosion, wear and fatigue resistance of the machined product [1,2]. In [3] it has been investigated how the stagnation point affects the residual stresses of the machined surface. In the studies presented in [4–6] the effect micro geometry of the insert and cutting speed has on the stagnation zone has been investigated. The aim of this paper is to investigate how the material flow at the stagnation zone is affected by the uncut chip thickness, which in turn will have an impact on the subsurface deformation of the workpiece.

Nomenclature

A	Johnson-Cook yield coefficient
a_p	Depth of cut
B	Johnson-Cook strain hardening coefficient
C	Johnson-Cook strain rate hardening coefficient
C_p	Specific heat capacity
E	Young's modulus
F_c	Primary cutting force
F_f	Feed force
h	Heat transfer coefficient
h_1	Uncut chip thickness
k	Thermal conductivity
n	Johnson-Cook strain hardening exponent
m	Johnson-Cook thermal softening exponent
P	Location of the stagnation point
q	Heat flux per unit of area
r_β	Edge radius
S	Separation location from machined surface and tool
T	Temperature
T_m	Melting temperature
T_X	Surface temperature in point X
T_Y	Surface temperature in point Y

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T_0	Bulk temperature
v_c	Cutting speed
x	Ploughed depth
α	Clearance angle
α_T	Thermal expansion
γ	Rake angle
ε	Strain
$\dot{\varepsilon}$	Strain rate
$\dot{\varepsilon}_0$	Reference strain rate
θ	Stagnation angle
κ	Major cutting angle
μ	Coulomb friction coefficient
ρ	Density
σ	Flow stress
σ_n	Surface normal stress
τ_f	Frictional stress
τ_y	Shear strength

2. Machining mechanics and theoretical aspects

This part of the paper covers the theoretical aspects of the machining process that will be investigated by the numerical simulations.

2.1. Material flow

In a cutting process, a stagnation zone is formed at the interface between the cutting tool and the workpiece. Since the material flow in the workpiece must either move towards the rake face of the cutting tool and form the chip or be ploughed under the cutting tool and turn into the machined surface. Fig. 1 illustrates the material flow in the workpiece in a cutting operation. Point *P* is the location of the stagnation point. Point *S* is the location where the machined surface separates from the tool. The ploughed depth *x* is a small fraction of the uncut chip thickness h_1. The line *PU* represents the shear plane. The vectors $\overrightarrow{OS}$ and $\overrightarrow{OP}$ has a length equal to the edge radius r_β. The stagnation angle θ, is given by the angle between the vectors $\overrightarrow{OS}$ and $\overrightarrow{OP}$ as given in Eq. 1.

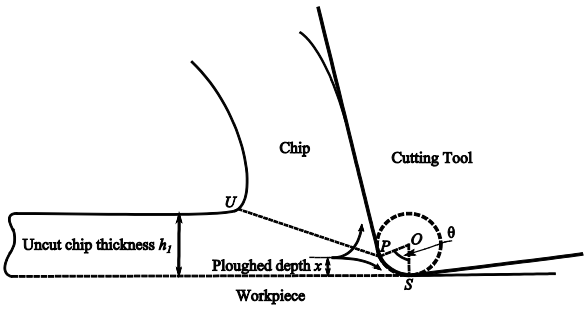


Fig. 1. Schematic view of the material flow.

$$\cos\theta = \frac{\left\langle \overrightarrow{OS}, \overrightarrow{OP} \right\rangle}{\| \overrightarrow{OS} \| \cdot \| \overrightarrow{OP} \|} \quad (1)$$

2.2. Stagnation Zone

Around point *P*, a stagnation zone will form where the material flow velocity of the workpiece will drastically decrease; thus, the shear stresses on the cutting tool will be close to zero in this region. The shear force component acting on the cutting tool, will change direction in the center of the stagnation zone at point *P*, illustrated by Fig. 2.

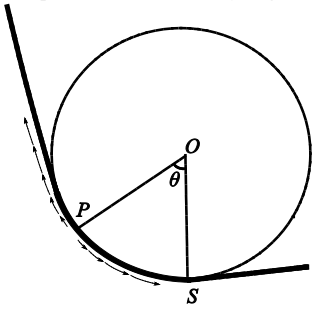


Fig. 2. Schematic view of the shear forces acting on the cutting tool at the stagnation zone.

3. Experimental investigation

As part of this investigation, experiments were performed through longitudinal turning of AISI 316L, with dry cutting conditions in a SMT 500 SWEDTURN CNC lathe. The tubes had an initial outer diameter of 80 mm and an inner diameter of 55 mm. The inserts used were industry standard cemented carbide inserts, SECO TOOLS CNMG120408-MF4 TM2000, with an edge radius r_β of approximately 45±5 µm. These inserts were mounted in a DCLNL 3225P12 tool holder with a major cutting angle κ=95° for all machining experiments. The cutting forces was measured with a cutting force sensor of the fabricate Kistler Z15814. The same cutting data was used for all the experiments; cutting speed v_c=125 m/min, while varying the feed in the steps 0.1 mm rev^{-1}, 0.2 mm rev^{-1}, 0.3 mm rev^{-1} and depth of cut a_p= 2 mm.

4. Numerical Modelling

4.1. FE-Formulation

The updated-Lagrangian formulation has been used for this analysis. Since Abaqus does not have a feature that automatically update the mesh, it has been performed with an implemented in-house Python script. The data transfer between the old and updated mesh has been achieved with the function *MAP SOLUTION. Fig. 3 shows a flow chart of the remeshing algorithm.

4.2. Material Modelling

The thermo-mechanical properties of both the tool and the workpiece are presented in Table 1. The temperature dependence of Young's modulus, specific heat capacity and thermal conductivity for the workpiece has been considered and is presented in Table 2. To model the flow stress of AISI 316L the Johnson-Cook plasticity model was used, Eq. 2.

Table 1. Thermo-mechanical material properties.

Material parameter	*AISI 316L*	*Cemented Carbide*
Density ρ [kg/m^3]	7900	15000
Thermal expansion α_T [µm/(mK)]	16.0	4.7
Thermal conductivity k [W/(mK)]	-	46
Specific heat capacity C_p [J/(kgK)]	-	203
Melting temperature T_m [K]	1672	-

Table 2. Temperature dependent material properties of AISI 316L [7].

Temperature T [K]	300	420	530	620	700	750	810	920	1470
Young's modulus E [GPa]	210	192	180	191	188	186	157	114	68
Thermal conductivity k [W/(mK)]	$k(T)$=14.307+0.0181·(T-273)-6E-6·(T-273)^2								
Specific heat capacity C_p [J/(kgK)]	$C_p(T)$=440.79+0.5807·(T-273)-0.001·(T-273)^2+7E-7·(T-273)^3								

$$\sigma = \left(A + B\varepsilon^n\right)\left(1 + C\ln\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)\right)\left(1 - \left(\frac{T - T_0}{T_m - T_0}\right)^m\right) \quad (2)$$

where ε is the plastic strain, $\dot{\varepsilon}$ is the strain rate, $\dot{\varepsilon}_0$ is the reference plastic strain rate, T is the temperature of the workpiece material, T_m is the melting temperature of the workpiece material and T_0 is the bulk temperature. Coefficient A is the yield strength, B is the hardening modulus, C is the strain rate sensitivity coefficient, n is the hardening exponent and m the thermal softening exponent.

The material constants used for modelling the plastic behavior of AISI 316L can be seen in Table 3.

Table 3. Johnson–Cook constitutive model constants.

A [MPa]	B [MPa]	C	n	m	$\dot{\varepsilon}_0$
305	441	0.057	0.10	1.041	1

In this study, the combination of the Coulomb's and shear friction model has been used, Eq. 3.

$$\tau_f = \min\left(\mu\sigma_n, \tau_y\right) \quad (3)$$

where τ_f is the frictional stress, μ is the friction coefficient, σ_n is the normal stress and τ_y is the shear strength of the workpiece, which is considered as $\tau_y = \sigma_y / \sqrt{3}$ where σ_y is the uniaxial yield stress. The reason this model is used is that the mechanism behind the frictional behavior is not fully understood, this model has been used for its simplicity.

There are mainly two sources of heat generation in a cutting process, plastic deformation and frictional work. 90% of the plastic deformation energy is considered to convert into heat. 100% of the frictional work at the contact interface is converted into heat and divided equally between the two bodies. The heat transfer coefficient, h between the tool and the workpiece is set to 1000 kW/m^{2}K. The conductive heat transfer between the contact surfaces is defined in Eq. 4.

$$q = h\left(T_X - T_Y\right) \quad (4)$$

4.3. Computational parameters

The cutting process was modelled as a two-dimensional model in Abaqus/Standard v.6.12-3. The time integration of the model was an implicit scheme. The cutting tool had a clearance angle α of 5°, a rake angle γ of 0° and an edge radius r_β of 50µm. The cutting speed was set to 125 m/min. Three simulations were performed with an uncut chip thickness, h_1 equal to 0.1, 0.2 and 0.3mm respectively. The elements used for this analysis were a 3-node plane strain thermally coupled triangle, linear displacement and temperature, (CPE3T in Abaqus). To capture the change in stagnation angle due to an increased uncut chip thickness, a very small element size is needed in both the workpiece and the cutting tool, at the area around edge radius. The element size in the workpiece, as a function of depth below the machined surface is presented in Table 4.

The mechanical boundary conditions of the FE-model are presented in Fig. 4. The back of the cutting tool is considered to be clamped. The bottom of the workpiece is considered to be fixed in all directions expect in the x-direction according to Fig. 4, where it has a velocity equal to the cutting speed. The thermal boundary condition at the interaction interface between the workpiece and the cutting tool is given by Eq. 4. No other thermal boundary conditions are present in the model since the heat transfer to the surrounding environment is neglected.

Table 4. The mesh gradient below the machined surface.

Distance from surface [µm]	Element side length [µm]
0	0.2
50	1.1
100	2.0
150	4.8
200	7.7
300	17.5
⋮	⋮
1900	250

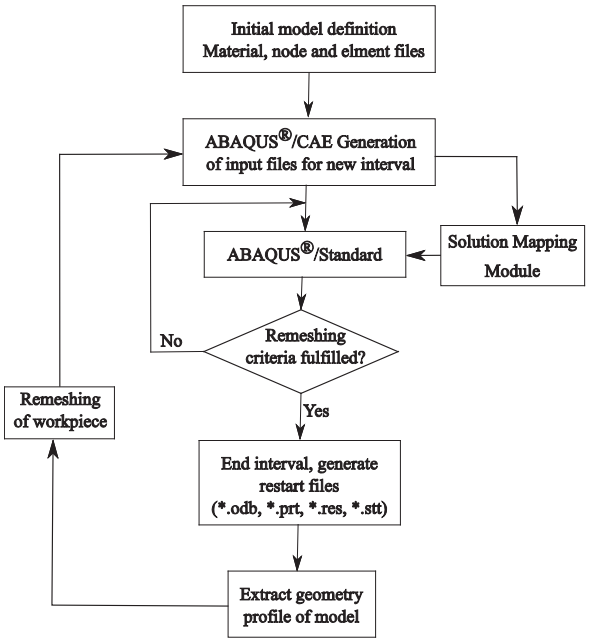


Fig. 3. Flow chart of the remeshing algorithm and overall simulation approach.

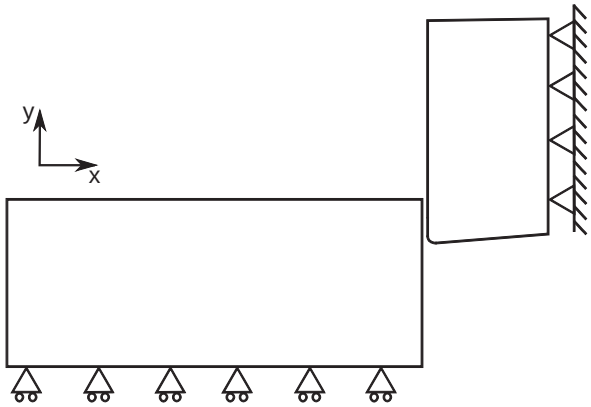


Fig. 4. Mechanical boundary conditions applied in the FE model.

5. Results

5.1. Cutting Forces

To be able to compare the experimental and simulated cutting forces. The uncut chip thickness has been converted to orthogonal cutting using Woxéns equivalent chip thickness h_e, described in [8]. In Table. 5, the correlation between the simulated and experimental cutting forces at the stationary phase is presented. As seen in Table 5, the agreements of the simulated and experimental values for both the primary cutting force and the feed force, are in the range of ±9%. This indicates that the FEM model can produce reliable results for contact forces and material flow of the cutting process.

Table 5. Comparison between experimental and numerical cutting forces.

	Experimental		*FEM*		*Error*	
h_1 (mm)	F_c (N)	F_f (N)	F_c (N)	F_f (N)	F_c	F_f
0.10	220(±6)	125(±4)	201	115	-8.8%	-7.8%
0.20	344(±11)	131(±5)	343	139	-0.3%	6.1%
0.30	482(±9)	153(±6)	499	165	3.5%	8.1%

5.2. Stagnation Zone

The material flow of the workpiece is presented in Fig. 5, where the field illustrate the node displacements in the workpiece. The stagnation zone is the blue area around the edge radius of the cutting tool. It has been shown experimentally by [5,9] that this is the location where the stagnation zone will form when cutting with a micro-geometry similar to the one used in this present simulation. The axial frictional force component acting on the tool is shown in Fig. 6, as seen there the axial frictional force component change direction, which indicate the location of the stagnation point *P*.

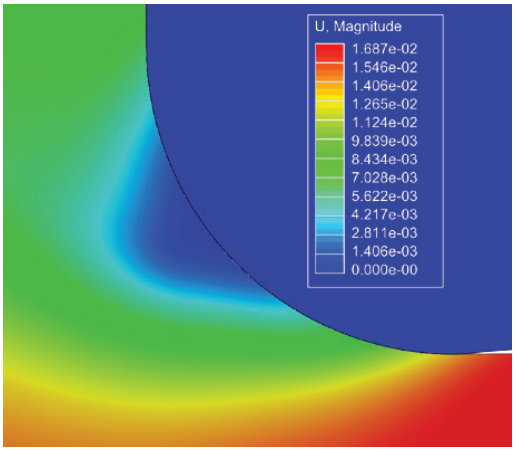


Fig. 5. The magnitude of displacement of the workpiece within a remeshing interval, with h_1=0.30 mm.

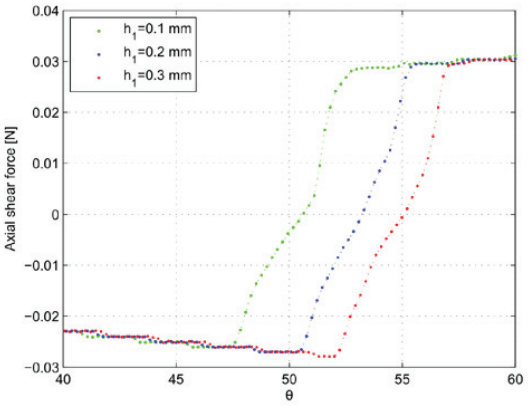


Fig. 6. Axial frictional force component acting on the tool along the edge radius.

5.3. Deformation Zone III

The deformation in the subsurface of a machined component, has a connection to the material flow in the cutting operation. In Fig. 7 the equivalent plastic strain in the subsurface is illustrated. Both magnitude of equivalent plastic strain and the deformation width are affected by an increase in uncut chip thickness. Because of this increased plastic strain in the subsurface the temperature affected zone is also increased as seen in Fig. 8. Fig. 9, shows that the material flow of the workpiece also has an impact on the stress in the tangential direction of the subsurface. The material affected by stresses in the tangential direction goes much deeper into the subsurface than the effect by plastic strain and temperature. Although as a clarification, the stress distribution shown in Fig. 9 is not the residual stress, it is the stress in the material just after the cutting tool has passed over the surface.

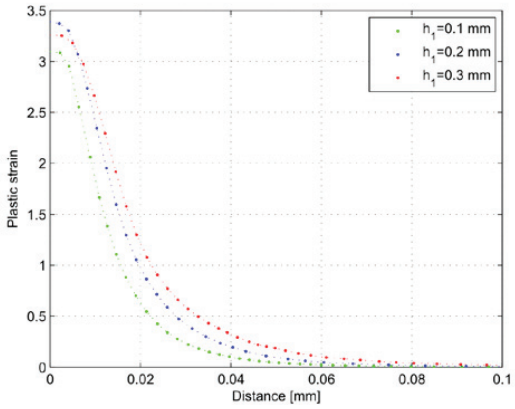


Fig. 7. Equivalent plastic strain in the machined surface.

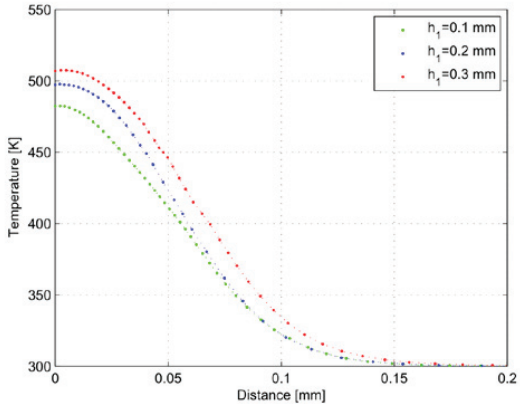


Fig. 8. Temperature in the machined surface.

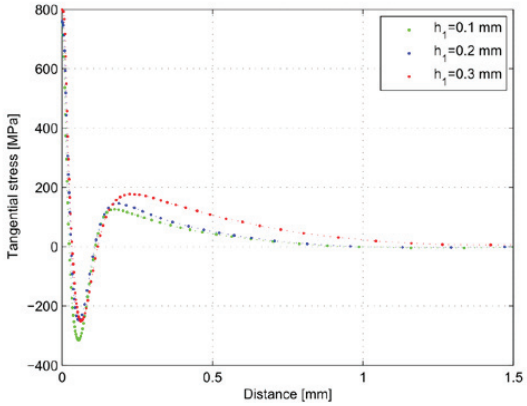


Fig. 9. Tangential stress component in the machined surface.

6. Conclusions

An increase in h_1 leads to a greater stagnation angle, which in turn force more material into the subsurface of the workpiece. This can be observed in Fig. 6, since the deformation hardened layer increase with h_1. A small h_1 give rise to higher comprehensive stress in the tangential direction. An increase of h_1 gives rise to higher tensile stresses in the machined surface at a depth greater 0.12 mm into the subsurface of the workpiece material.

One important aspect that has not been considered in this current research is that with a segmented chip, the stagnation point will probably oscillate around a specific interval of the stagnation angle, with the same frequency as the segmentation. Although the aim of this paper was to identify that the stagnation angle is affected by the uncut chip thickness and what impact it has on the subsurface of the workpiece. This is easier to observe for a continuous chip formation, since there the shear angle is static.

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